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Making Tennis Game Predictable: Uncovering the Secrets of momentum through Data Analysis

Summary

In sports, momentum is often a challenging phenomenon to measure, but beyond the athletes' individual skills, it is also a crucial factor influencing the outcome of a competition. This has sparked contemplation on how momentum is generated or altered through a series of events during a match and how it manifests in sports competitions. The momentum of athletes is paramount for winning a game, as it provides valuable information for assessing their condition, predicting athlete data, and making timely adjustments. To better analyze athlete momentum and offer improved recommendations during a match, we have conducted in-depth research on this topic from various perspectives and levels.

Firstly, to better capture the major influencing factors and assess the momentum of players, we **preprocessed the data**. After an extensive literature review, we removed factors that insignificantly impact momentum and match results. We **validated the data's reasonability** through Python, checking for its completeness and handling entries with severe missing data in a special manner. To ensure the accuracy of subsequent computations, we **repaired missing and abnormal values** using the median. Additionally, to better evaluate the impact on momentum in subsequent analyses, we defined variables such as **relative momentum.**

To quantify the momentum of players, we established a mathematical model to normalize the selected data items. We applied the **entropy weighting** method to assign weights, and finally incorporated these weights into the **TOPSIS algorithm** to calculate the momentum. The progression of the matches was visualized using line graphs and scatter plots, with the graphical trends aligning with the course of the matches. We observed that momentum exhibited similarity with consistently successful patterns in the visual representations. In the subsequent sections, we further validated the similarity between the two.

To investigate the relationship between momentum and match outcomes, we conducted a correlation analysis from two dimensions: in-game momentum with consecutive scoring and total match momentum with the match results, using **Spearman's rank** correlation coefficient. The correlation coefficients were **0.641** and **0.674**, respectively. Subsequently, we assessed the significance of our model based on the **Z statistic**, which showed **a strong correlation** between these two factors.

In question three, in order to predict the course of the match based on historical data, we **defined "tp"** (turning points), where the occurrence of a turning point signifies a change in the game's direction. We employed **GBDT classification** to predict the match by identifying these turning points. Additionally, we delved deeper into the data features of turning points, analyzing their significance from the perspective of player 1. This analysis provided insights into the quality of the turning points, enabling us to offer recommendations to player 1, such as reducing running distance, to maintain an advantage and avoid entering a disadvantaged position.

We also used the model derived from the third question to test other matches we participated in, of which three had a prediction accuracy of over **80%.** However, a certain deviation occurred in one match. We learned from the influences we had not considered before and made improvements to the model, further improving its predictive accuracy. We also used this model to predict women's tennis and table tennis matches. The results showed good predictions for women's tennis matches, but the predictions were less satisfactory for table tennis matches due to the significant differences in competition format and influencing factors compared to tennis. This indicates that if we want to improve the accuracy of its forecasts, we should focus on reselecting and analyzing the feature indices.

Keywords: monument; GBDT Classification; TOPSIS algorithm

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Introduction

1.1 Problem Background

In the men's singles final at Wimbledon in 2023, 20-year-old Spanish newcomer Carlos Alcaraz defeated Grand Slam champion Novak Djokovic to claim the title. In the match, the incredible turnarounds often observed on a seemingly advantaged player were attributed to "momentum." Not only in tennis, but in various competitions, we often describe a player who scores consecutively as "unstoppable," indicating that momentum not only reflects the player's current performance state but also to some extent predicts the course of the game.

A player controls the rhythm of the entire tennis match, seemingly heading towards victory effortlessly, and then suddenly, their opponent gets an unexpected break or break point opportunity, and the momentum starts to reverse. The entire situation could potentially change completely at this pivotal moment. These decisive moments are often limited to key break or important hold games, but in reality, the key points leading to these situations are equally important[1].

Modern tennis players have strong creative abilities, but the ability to seize opportunities and turn advantages into winning opportunities is the fundamental difference between top athletes and ordinary players[2]. If we can quantify and grasp the momentum indicator, we can make decisive adjustments in tactics at critical junctures, thus getting closer to victory[3].

1.2 Restatement of the Problem

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

Question 1: Develop a model to capture the match flow at the occurrence of points and assess the performance of each player at that moment, along with their relative performance. Validate the evaluative model and apply it to a specific match process in a visualized manner.

Question 2: Establish a model to determine whether there is a causal relationship between fluctuations in a player's momentum and the continuity of success. Validate the accuracy of the results.

Question 3:

1.Develop a mathematical model to predict momentum fluctuations during a match and identify the most influential factors affecting the game.

2.Based on the differences in momentum changes in past matches, provide recommendations for a player facing different opponents in a new match.

Question 4: Test the fitting degree of the predictive model against actual results and its applicability to other matches. Identify factors that may need to be incorporated.

1.3 Our Work

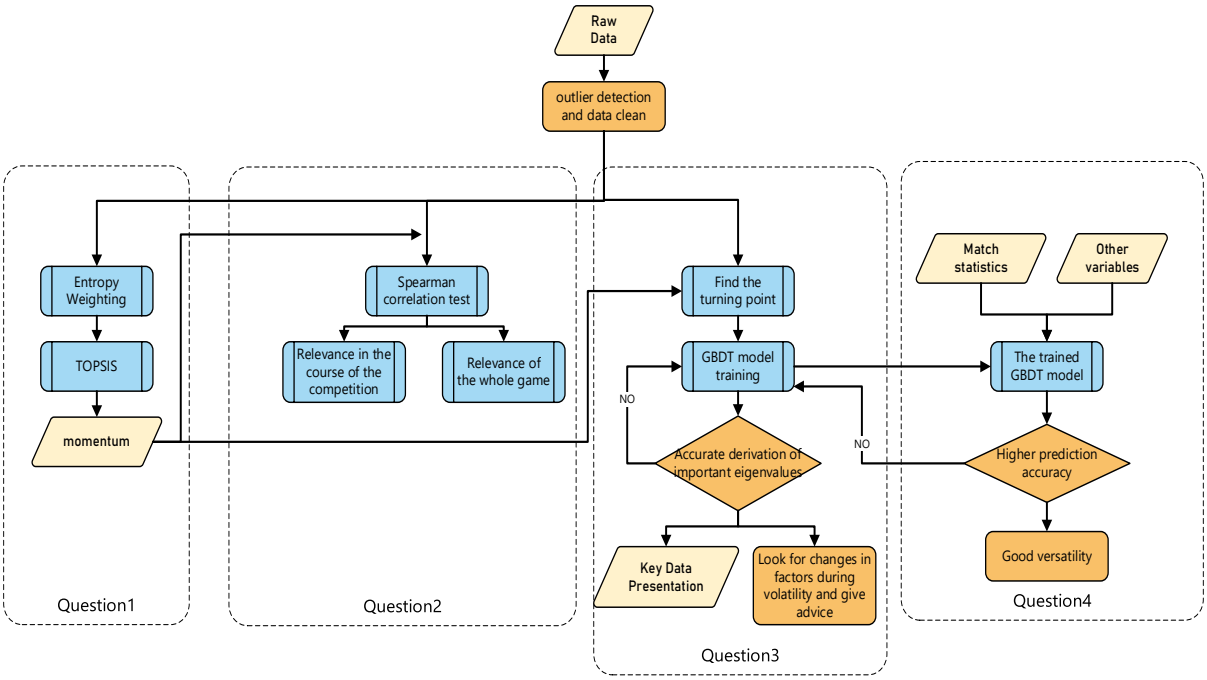

Figure 1: **Our Work**

Figure 1 is a concise flowchart outlining the structure of this paper. We have segmented the presentation of solutions to the four issues in the case study, but these four problems are not isolated; instead, they are interconnected. Subsequent questions will utilize data from previous ones.

2 Assumptions

To simplify the problem, we make the following basic assumptions, each of which is properly justified.

1.The provided tennis match data is sufficient to reflect the actual situation on the court.

2.The skill levels of players remain constant during the match, unaffected by factors such as fatigue, injuries, or psychological pressure. This assumption helps to simplify the model, making it easier to handle.

3.All players are solely focused on winning, without considering the influence of external factors such as gambling.

4.The current state of the match at a given moment depends only on the preceding moment's state and is independent of past match states.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description	Lable
W_{ai}	player i hit an untouchable winning serve（i = 0,1,2)	pi_ace
W_{wi}	player i hit an untouchable winning shot（i = 0,1,2)	pi_winner
W_{bi}	player i has an opportunity to win a game another player's serving（i = 0,1,2)	pi_break_pt
W_{pi}	player i won the game another player is serving(i = 0,1,2)	pi_break_pt_won
W_{di}	player i missed both serves and lost the point(i = 0,1,2)	pi_double_fault
W_{ui}	player i made an unforced error(i = 0,1,2)	pi_unf_err
WS	The number of consecutive wins	winner_stream
SM	speed of serve (miles per hour; mph)	speed_mph
SE	server of the point	server
AD	The difference in distance between the two players in each points	ave_dis
RC	number of shots during the point	raly_count
M_i	Momentum of the player i	pi_momentum
RM	Difference in momentum of player 1 and player 2	relative_mom
RMW	The sum of relative_mom in the whole game	relative_mom in a whole game
O	outcome of the competition	outcome

4 Data Preprocessing

After extensive review of relevant literature, we have identified the key winning factors in tennis, which include ACEs, double faults, first and second serve effectiveness, winners, and unforced errors, as highlighted in the research by Ma Jiacai[4], Wang Qianru[5], and the official indicators published by the Sports Performance Analytics (SPA) organization[6]. However, due to limited data availability, it is challenging to calculate the rates of various player statistics, and overall rates may not accurately reflect the momentum of players at any given moment. Therefore, this paper treats the occurrence of these events as qualitative indicators rather than considering the probability of their occurrence.Finally, we extract,speed_mph, break_pt_missed,rally_count,count1, server, ace, break_pt,winner,unf_err,break_pt_won, double_fault as the primary indicator in the model.

We processed the "point_victor" in the dataset using MATLAB to calculate the consecutive winning streaks, referred to as "winner-stream." We defined the average running distance difference as "ave_dis." This was computed by subtracting "p2_distance_run" from "p1_distance_run" and dividing the result by "rally_count." For cases where "rally_count" is 0, we

explicitly set the value of "ave_dis" to 0.

During the preliminary analysis of the provided dataset, we utilized Python to examine missing values and outliers. By defining the range of values for continuous data and specifying legitimate values for discrete data, we checked for the presence of null values or abnormally large values.

We identified significant data gaps in the 'speed-mph' and 'winner-shot-type' columns during a specific match. Upon further investigation, we observed that, in cases of serving failures, the shot count was consistently zero, 'speed-mph' was marked as NA, and the characteristics of these indicators for player1/2 at the respective scoring points were identical, showing no variability.

A few other columns also exhibited minor instances of missing or outlier values. We removed outlier values, converting them into null values, and filled these missing values with the median of the respective indicators. Specifically, we replaced missing values in the 'speed-mph' and 'winner-shot-type' columns with zeros. Additionally, using the entropy weighting method, we initially calculated the weights of each influencing factor.

Table 2: The weight table before adjustments.

items	information entropy e	information utility value d	weights (%)
p1_ace	0.66	0.327	20.861
p1_winner	0.802	0.198	13.276
p1_break_pt	0.635	0.315	19.501
p1_break_pt_won	0.527	0.373	24.071
winner_stream	0.998	0.12	7.815
speed_mph	0.987	0.013	0.874
server	0.92	0.142	10.385
p1_double_fault	0.998	0.032	2.127
p1_unf_err	0.985	0.015	1.019
ave_dis	1	0	0.026
rally_count	0.999	0.001	0.045

When applying the entropy weight method to determine weights based on the non-missing parts of these two columns, it was observed that their direct impact on the competition advantage level (momentum) is relatively low. Additionally, the excessive number of feature indicators can compromise the stability of the entropy weight method[7], thereby diminishing the scoring effectiveness of the TOPSIS model. For these specific indicators, in the calculation of momentum in the first question, we temporarily exclude the consideration of the features that are missing.

5 Evaluate player based on momentum

The playing field is constantly shifting. Every moment of each match consists of numerous factors that influence players' performance, which in turn affects the outcome of the match.

Analyzing these changing data can reflect a player's performance at a certain point to some extent. By studying the data and trends of players' performance throughout the match, we can assess the overall skill level of the players and their performance at different stages, and guide their tactics in future matches. Therefore, in this section, we apply the TOPSIS algorithm to evaluate the players and use the entropy weight method to determine the weight of each factor. We carried out tests using the provided match data, and finally used the match between Carlos Alcaraz and Novak Djokovic as an example to compare the performances of the two players.

5.1 TOPSIS-Entropy Weights Method

TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) is a multi-criteria decision-making method developed in the field of operational research. It involves numerical values and it can be used for ranking multiple alternatives based on several attributes or criteria.

The main idea behind the TOPSIS algorithm is to identify an ideal solution (representing maximum benefits) and a negative-ideal solution (representing minimum rewards). All alternatives are then ranked based on their proximity to the ideal solution and distance from the negative-ideal solution. The alternatives are evaluated on a range of criteria, which can be assigned different weights based on their importance.

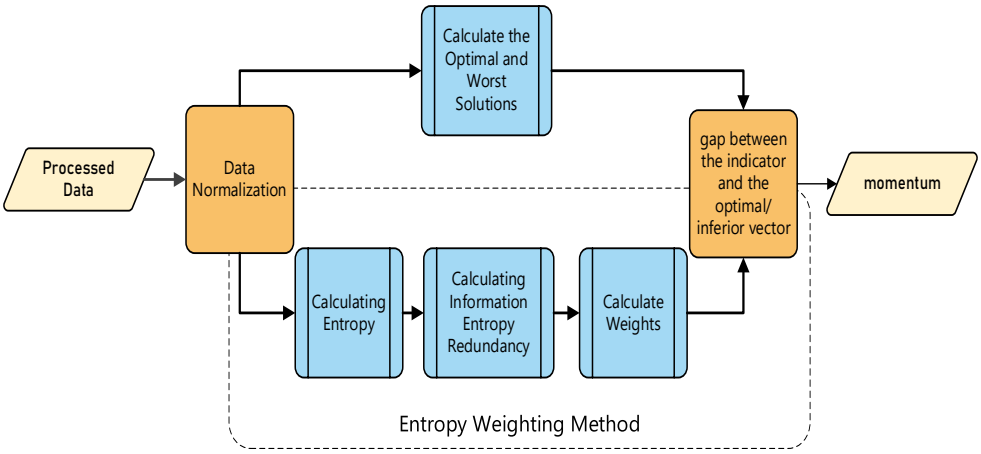

Figure 2: **Flowchart for question one**

Data Normalization

$$X_{\min} = \min(X_{1j}, X_{2j}, \ldots, X_{nj}) - 0.0001 \quad (1).$$

$$X_{\max} = \max(X_{1j}, X_{2j}, \ldots, X_{nj}) + 0.0001 \quad (2).$$

For positive factors:

$$z_{ij} = \frac{X_{ij} - X_{\min}}{X_{\max} - X_{\min}} \quad (3)$$

For negative factors:

$$z_{ij} = \frac{X_{\max} - X_{ij}}{X_{\max} - X_{\min}} \quad (4)$$

This article undertakes normalization and harmonization processes. The offset values in formulas (1) and (2) are employed to ensure that there are no instances of a denominator being zero in subsequent calculations.

In the two formulas (3) and (4), z_{ij} represents the standardized values for positive and negative indicators, respectively. In the calculation of positive indicators, the data is normalized to the $[0,1]$ range by subtracting the minimum value $X_{\min}$ from X_{ij} and then dividing by the difference between the maximum and minimum values. Similarly, in the calculation of negative indicators, the data is normalized to the $[0,1]$ range by subtracting X_{ij} from the maximum value $X_{\max}$ and then dividing by the difference between the maximum and minimum values.

Weighting Calculation

$$p_{ij} = \frac{z_{ij}}{\sum_{j=1}^{m} z_{ij}} \tag{5}$$

$$e_j = -k\sum_{i=1}^{n} p_{ij}\ln(p_{ij}), \quad j=1,\cdots,m$$
$$k = 1/\ln(n) > 0, e_j \geq 0 \tag{6}$$

$$d_j = 1 - e_j, \quad j=1,\cdots,m \tag{7}$$

$$w_j = \frac{d_j}{\sum_{j=1}^{m} d_j}, \quad j=1,\cdots,m \tag{8}$$

In formula (5), m represents the number of indicators, and p_{ij} denotes the weight proportion of the j-th indicator during the i-th scoring.

Formula (8) normalizes the calculated weights, ensuring that the sum of all weights is equal to 1.

The entropy method, based on the definition of information entropy, uses entropy values to assess the degree of dispersion for a particular indicator. The smaller the information entropy value, the greater the indicator's dispersion, indicating a stronger impact on the overall evaluation. If the values of a specific indicator are all equal, that indicator has no effect on the comprehensive evaluation. Therefore, by utilizing the tool of information entropy, weights for each indicator can be computed, providing a basis for the comprehensive evaluation of multiple criteria.

Calculating Momentum

$$(Z_1^+, Z_2^+, ..., Z_m^+) \tag{9}$$
$$(Z_1^-, Z_2^-, ..., Z_m^-) \tag{10}$$

$$D_i^+ = \sqrt{\sum_{j=1}^{m} w_j (Z_j^+ - z_{ij})^2}, \quad D_i^- = \sqrt{\sum_{j=1}^{m} w_j (Z_j^- - z_{ij})^2} \tag{11}$$

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-} \tag{12}$$

First, we identified the optimal and worst matrix vectors (8) and (9). Subsequently, we computed the values D^+ and D^-, where these values represent the distance between the evaluated object and the optimal or worst solution. D^+ larger value indicates a greater distance, with a higher D^+ value implying a greater distance from the optimal solution and a higher D^- value indicating a greater distance from the worst solution. The ideal research object is one with a smaller D^+ value and a larger D^- value.

The comprehensive score, denoted as C value, is calculated as $C = (D^-)/(D^+ + D^-)$. In the formula, the numerator represents the D^- value, and the denominator is the sum of D^+ and D^-. A relatively larger D^- value suggests that the research object is farther from the worst solution, indicating better performance. A larger C value implies a better research object.

5.2 Visual Representation of the Flow of Play

Table 3: Modified Weights

items	information entropy e	information utility value d	weights (%)
p1_ace	0.66	0.327	21.079
p1_winner	0.802	0.198	13.403
p1_break_pt	0.635	0.315	19.735
p1_break_pt_won	0.527	0.373	24.074
count1	0.998	0.12	8.116
server	0.92	0.142	10.436
p1_double_fault	0.998	0.032	2.129
p1_unf_err	0.985	0.015	1.028

As shown in Table 3, after removing some irrelevant indicators during the data preprocessing stage, we recalculated the weights of each indicator using the entropy weight method. Subsequently, we applied the TOPSIS model to determine the momentum of players at each scoring point, and ranked the momentum at different time points. Table 4 displays the momentum and corresponding rankings at selected scoring points.

Table 4: Momentum and ranking

point_no	ositive Ideal Solution Distance	Negative Ideal Solution Distance	Combined Score Index	ranking
1	0.96548705	0.24558572	0.20278362	736
2	0.96022837	0.26540357	0.21654427	563
3	0.88886094	0.44989311	0.3360536	281
4	0.75018615	0.65551018	0.46632417	119
5	0.75022236	0.65547234	0.4662978	120
6	0.96022837	0.26540357	0.21654427	563
7	0.96020009	0.26549699	0.21660898	562
8	0.98755879	0.13166615	0.11764047	960

We selected the match between Carlos Alcaraz and Novak Djokovic as an example to calculate the momentum throughout the entire game. We visualized the entire match process using a line chart, where Carlos Alcaraz is represented as player1 and Novak Djokovic as player2. The peaks of the yellow line in Figure 3 indicate that, in the early stages of the match, Novak Djokovic had a stronger momentum, while the peaks of the blue line, although smaller, are more concentrated between the 50th and 75th points, indicating that player1 was not at a disadvantage. This aligns with the result of him losing the second set with a narrow 7-6 score to player2. In the third set, player2 showed a clear decline, with only two instances of momentum exceeding 0.5 and three instances exceeding 0.4.

By observing the two scatter plots on the right side of Figure 3, we can see that in the first half of the match, player1 had 8 instances of scoring more than 3 consecutive points, while player2 had 11 instances. The instances of scoring more than 2 consecutive points were 16 for player1 and 22 for player2. This suggests that player2 had a stronger scoring ability in the first half of the match. However, in the second half of the match, player1 had significantly fewer instances of scoring consecutively compared to player2. Therefore, we can speculate that in the latter part of the match, player1 surged ahead continuously and secured the victory, which indeed turned out to be the case.

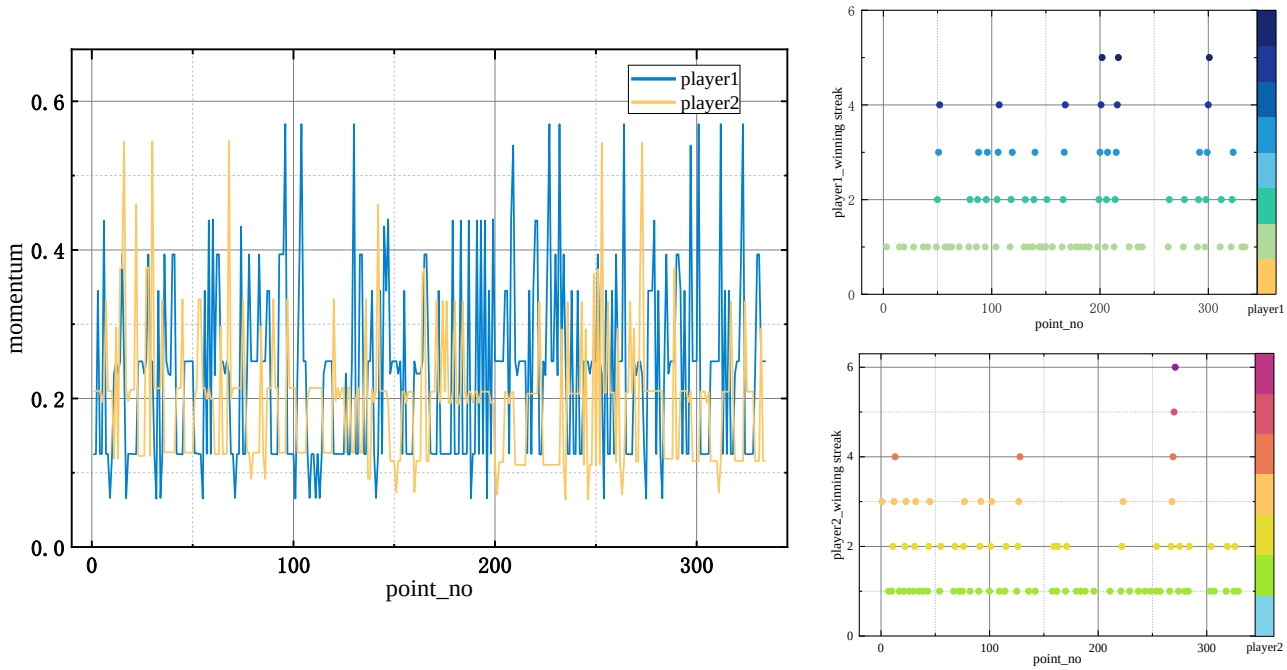

Figure 3: **Visualization of the competition process**

Here, we define the "player_better point," stipulating that if any one or more of the following events occur - "p1_ace," "p1_winner," "p1_break_pt," or "p1_break_pt_won," the point is considered a "p1_better point." Since these four indicators are all positive and carry significant weight in the TOPSIS model, we believe they effectively reflect the player's outstanding

performance at that specific scoring point. Through observation, we note that the occurrence of higher points aligns with the peak positions in the player's momentum on the line chart. This similarity indicates crucial information points in the flow of the match.

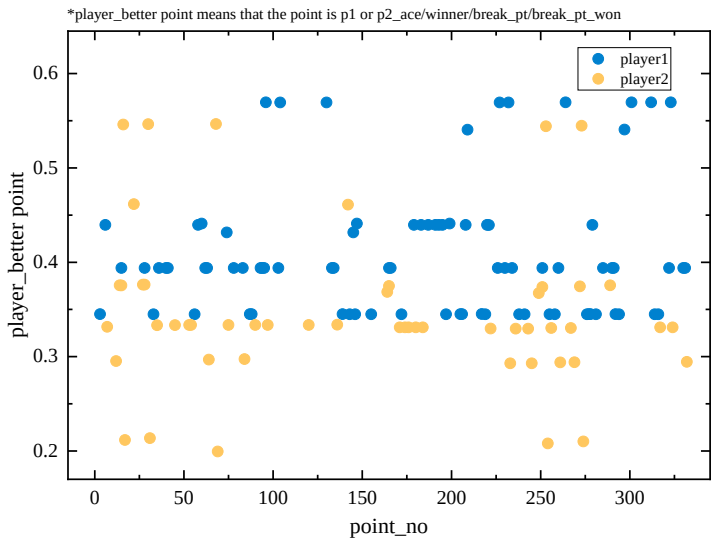

Figure 4: **Better-point**

6 Does the fluctuation of momentum affect runs of success?

We conducted a correlation analysis on the momentum and scoring ability of players at certain points in the game, as well as the overall game momentum and outcomes, using Spearman's rank correlation coefficient and significance testing (Z-test) to investigate whether momentum affects continuous scoring and ultimately the result of the game.

6.1 Spearman's Correlation Analysis

The Spearman correlation coefficient is widely used in social sciences, behavioral sciences, and other fields as a non-parametric measure to assess the degree of association between the ranks of two variables. Unlike the Pearson correlation coefficient, the Spearman correlation coefficient does not assume that the data are normally distributed, nor does it assume a linear relationship between the two variables. Therefore, it is suitable for evaluating nonlinear relationships as well as the relationships between ordinal data or interval/ratio data that do not fully meet the normal distribution assumption. The value of the Spearman correlation coefficient ranges between -1 and 1, where 1 indicates a perfect positive correlation, -1 indicates a perfect negative correlation, and 0 indicates no correlation.

The steps to calculate the Spearman correlation coefficient are as follows:

1.Arrange the data for each variable in ascending order and assign the corresponding ranks.
2.Calculate the square of the difference in ranks for each pair of data.
3.Apply the formula for the Spearman correlation coefficient:

The Spearman correlation coefficient is defined as the Pearson correlation coefficient between ranked variables, where the original data are assigned ranks based on their average descending position within the total dataset. For a sample of size n, the n pieces of original data are converted into rank data, and the correlation coefficient ρ is calculated accordingly.

$$\rho=\frac{\sum_i(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_i(x_i-\bar{x})^2\sum_i(y_i-\bar{y})^2}} \tag{13}$$

In practical applications, the ties between variables are considered insignificant, thus simplifying the steps to calculate ρ Here, d_i is the difference between the ranks, and n is the number of data pairs.

$$\rho=1-\frac{6\sum d_i^2}{n(n^2-1)} \tag{14}$$

Here, we conduct a correlation analysis using Formula (14), calculating the value of the correlation coefficient ρ after hypothesizing. For larger samples ($n>30$), the distribution of Spearman's rank correlation coefficient can be approximated as a normal distribution. In such cases, the Z statistic can be used for hypothesis testing. The Z statistic is calculated by converting Spearman's rank correlation coefficient ρ into a Z score, with the conversion formula typically as follows:

$$\mathrm{Z}=\frac{\operatorname{arctanh}(\rho)}{\sqrt{\frac{1}{\mathrm{n}-3}}} \tag{15}$$

The calculated Z-value can be compared with the standard normal distribution to determine the significance of the correlation. Generally, we conduct hypothesis testing at a significance level of 0.05. The rejection region is $(Z<-1.96)\bigcup(Z>1.96)$.

6.2 Examination of the ability to score points during the course of the game

In the TOPSIS scoring calculation for momentum in the first question, we computed the momentum for both player1 and player2, denoted as player1_mom and player2_mom, respectively. Subsequently, we determined the relative momentum by taking the difference between the two, representing how each player's performance compares to the other at a specific scoring point (defaulting to player1 compared to player2). This relative momentum is denoted as "relative_mom." In the subsequent analysis, we compared the correlation between momentum and relative momentum with consecutive scores and found that relative momentum is more suitable for representing the performance between players.

Proposing the hypothesis:

1.For the correlation between indicators WS and M_i:

$H_0: \rho_{WS\,M_i} = 0$ The indicators WS and M_i are not correlated.

$H_1: \rho_{WS\,M_i} \neq 0$ The indicators WS and M_i are correlated.

2.For the correlation between indicators WS and RM:

$H_0: \rho_{WS\,RM} = 0$,The indicators WS and RM are not correlated.

$H_1: \rho_{WS\,RM} \neq 0$,The indicators WS and RM are correlated.

3.For the correlation between indicators M_i and RM:

$H_0: \rho_{M_i\,C} = 0$ The indicators M_i and RM are not correlated.

$H_1: \rho_{M_i\,C} \neq 0$ The indicators M_i and RM are correlated.

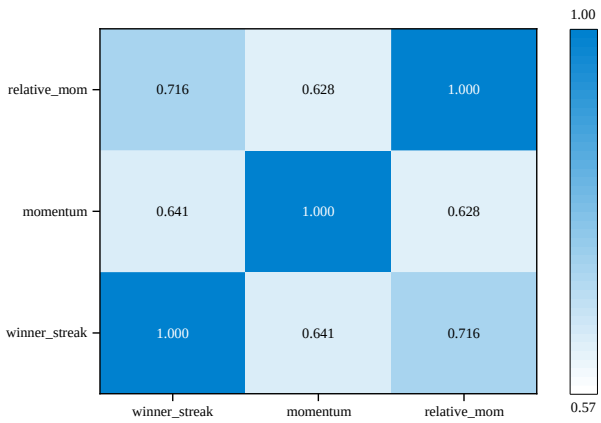

Figure 5: **In-Game Momentum - Winning Streak Heatmap**

After calculating the correlation coefficients for WS and M_i, WS and RM, M_i and RM, we obtained the correlation heatmap shown above. It can be observed that the correlation coefficient for WS and M_i is 0.641, for WS and RM is 0.716, and for M_i and RM is 0.628, indicating strong correlations among the three. Comparing the correlation coefficients between relative momentum and consecutive scoring ability with those of momentum and consecutive scoring ability, we found that, during the course of the game, relative momentum better represents a player's scoring ability at a given moment than momentum.

Upon further calculation of Z, we obtained Z values for WS and M_i as 64.6648, WS and RM as 76.5379, and M_i and RM as 62.8128. Clearly falling within the rejection region, we reject the null hypothesis for all three, suggesting the presence of a correlation among WS, M_i and RM.

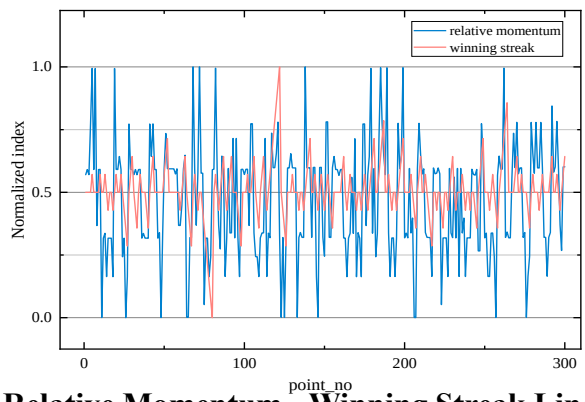

Figure 6: **Relative Momentum - Winning Streak Line Chart**

From Figure 6, it can be observed that the trend of consecutive victories aligns closely with that of relative momentum, with the peaks occurring almost simultaneously.

6.3 Tests for winning or losing the whole game

In the previous section, we concluded that at each scoring point in a game, the relative momentum is more correlated with consecutive points, x, than the momentum, indicating that relative momentum is more related to scoring ability. In this section, we introduce the total relative momentum of the entire game (the sum of the relative momentum at each scoring point in the game) and analyze its correlation with the game outcome. We continue to use Spearman's correlation coefficient and combine it with a Z-test with a significance level of 0.05

following hypotheses are proposed:

$H_0: \rho_{RMW\,O} = 0$ The indicators RMW and O are not correlated.

$H_1: \rho_{RMW\,O} \neq 0$ The indicators RMW and O are correlated.

After calculation, as shown in Figure 7, we found that the correlation coefficient between the total relative momentum and the game outcome is 0.674, which shows a strong positive correlation. The Z-test value is 69.6148, which falls within the rejection region. Therefore, we reject the null hypothesis and conclude that there is a correlation between the relative momentum and the game outcome.

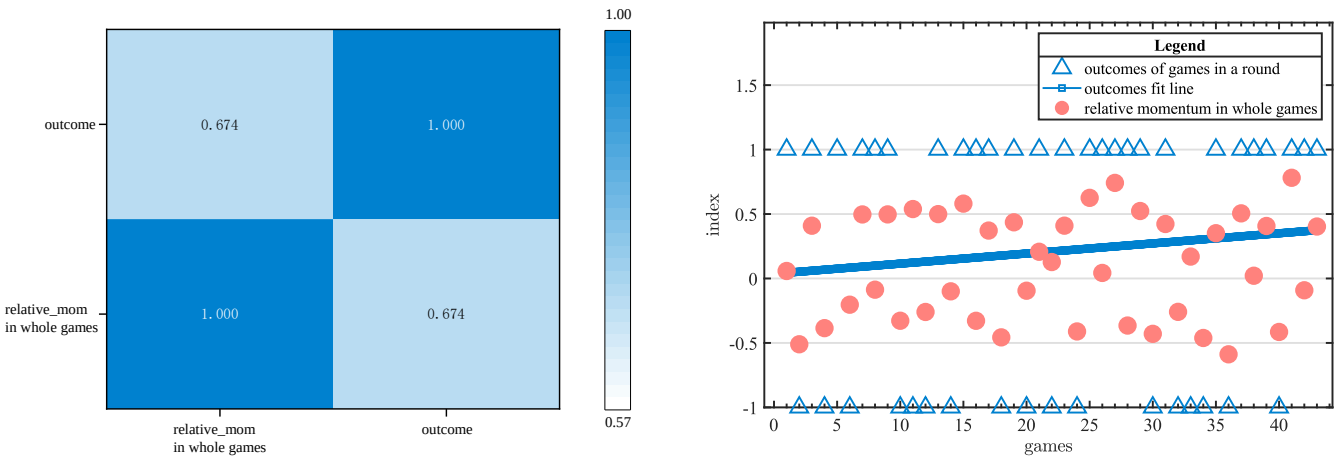

Figure 7: **Relative Momentum and Winning the Race**

At the same time, we perform a linear regression on the results of each game, obtaining a fitted curve, and drawing a scatter plot. In the graph, we find that the total relative momentum of the games is evenly distributed on both sides of the fitted curve of the game results. This suggests that the relative momentum and the game results share the same trend and can represent the outcome of the game quite accurately.

7 Predicting the course of the game and providing strategies

7.1 Game Trend Prediction Based on GBDT

7.1.1 Defining Turning Points

We calculate the difference series for p1 and p2, denoted as p1-ds and p2-ds, respectively, to show the momentum changes of p1 and p2. For p1-mom, it can be assumed that when p1-ds experiences a sign change, p1-mom fluctuates more significantly, indicating a turning point. Similarly, this rule also applies to p2 with p2-ds. For relative-mom, when it has a sign change, we also consider it a turning point. We define a scoring point as the turning point of the game, denoted as tp, when these three undergo a turning at the same time.

We normalize ralative-mom, p1-mom, and p2-mom, and plot the fluctuation curves of all three. The graph below illustrates the fluctuation curve concerning the scoring points in the same match, with the position of the turning points marked with brown dotted lines.

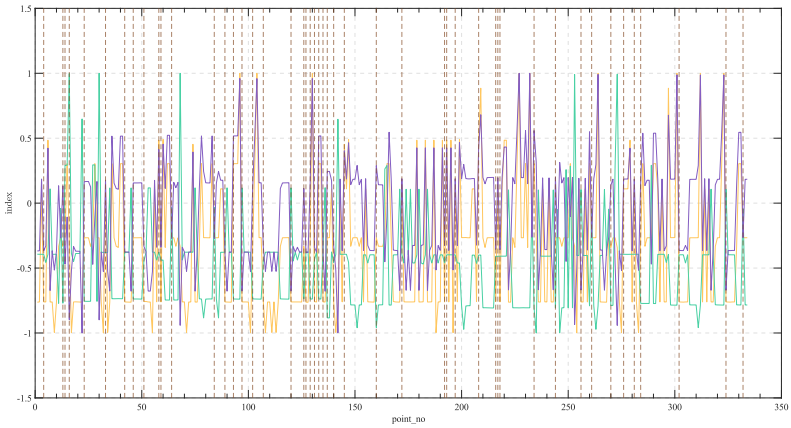

Figure 8: **Turning point**

After the turning points occur, the trend of the game changes, indicating that whenever there's a turning point, we can expect a turning and fluctuation in the match's pace. Therefore, at any scoring point, if we want to predict whether the future match will experience a turning or fluctuation, we only need to judge whether this point is a turning point.

7.1.2 GBDT Modle

The GBDT (Gradient Boosting Decision Tree) model is an additive model that sequentially trains a set of CART (Classification and Regression Trees) regression trees. The final prediction is obtained by summing the predictions of all the regression trees, resulting in a strong learner. Each new tree is fitted to the negative gradient direction of the current loss function. The output of this ensemble of regression trees is then processed through a sigmoid or softmax function to obtain binary or multiclass classification results. The core idea is to fit the negative gradient of the loss function to the residual of the previous round's base learner, gradually reducing the estimation of residuals in each round. Fitting in the negative gradient direction ensures that each round of training efficiently minimizes the loss function, accelerating convergence towards local or global optimal solutions[8].

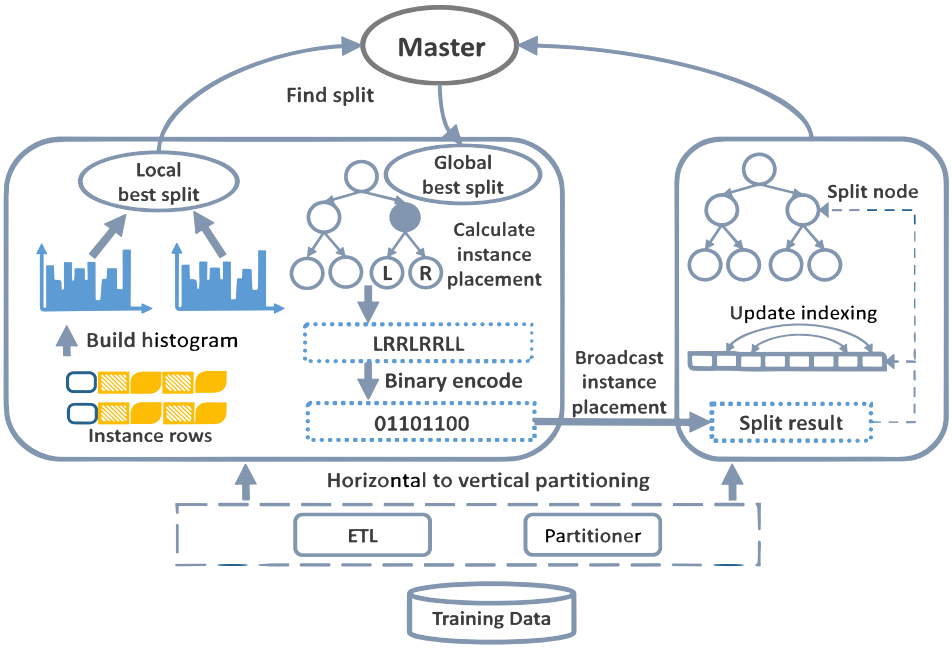

Figure 9: **GBDT flow chart**

The formula for GBDT is as follows:

（1）initialization

$$f_0(x) = \underset{\theta}{\operatorname{argmin}} \sum_{i=1}^{N} L\left(y_i;\theta\right) \tag{16}$$

(2) iteration: Perform the following steps for $m=1,2,\cdots,M$

①$i=1,\ 2,\ \cdots,N$ calcuate

$$r_{mi} = -\left[\frac{\partial L\left(y_i,f(x_i)\right)}{\partial f(x_i)}\right] f(x) = f_{m-1}(x) \tag{17}$$

That is, using the negative gradient of the loss function for each sample as an approximation of the sample residual $r_{mi} = y_i - f(x_i)$

② Fit a CART regression tree to r_{mi} to get the set of leaf nodes R_{mj}, $j=1,2,\ \cdots,\ J$。

③Loss function for minimization of $j=1,\ 2,\ \cdots,\ J$

$$\theta_{mj} = \underset{\theta}{\operatorname{argmin}} \sum_{x_i \in R_{mi}} L\left(y_i f_{m-1}(x_i) + \theta\right) \tag{18}$$

④Updated forecast results:

$$f_m(x) = f_{m-1}(x) + \sum_{j=1}^{J} \theta_{mj} l(x \in R_{mj}) \tag{19}$$

(3) Get the GBDT model.

$$\hat{f} = f_M(x) = \sum_{m=1}^{M} \sum_{j=1}^{J} \theta_{mj} I(x \in R_{mj}) \tag{20}$$

7.1.3 Training and testing

We selected match 1701 for training and testing, using 80% as the training group and 20% as the test group, conducting machine learning with 100 base learners and a learning rate of 0.1. The following is the variable selection:

Variable X: {ave-dis, speed-mph, rally-count, momplayer1, count1, relative-mom, momplayer2};

Variable Y: { turning point}

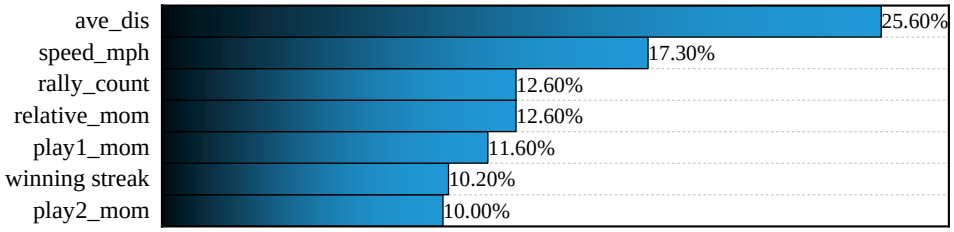

Figure 10: **importance of Indicator Characteristics**

Through the gradient boosting tree built to calculate feature importance, it was found that the importance of the above indicators all exceeded 10%. Among them, ave-dis and speed-mph were the most important, and there was not much difference in the importance of the other four.

Table 5: Projected Accuracy

	Accuracy	Recall	Precision	F1
Training Sets	1	1	1	1
Test set	0.851	0.851	0.851	0.851

The above table shows the prediction evaluation indicators of the cross-validation set, training set, and test set, used to quantify the prediction effect of GBDT. Among them, the evaluation indicators of the cross-validation set can be used to continuously adjust the hyperparameters to obtain a reliable and stable model. From the table, it can be seen that the four accuracy-describing indicators are all 0.851, indicating that the model has a high prediction accuracy and strong usability.

7.2 Recommendations by analyzing turning points

The figure below displays the frequency distribution histograms describing the data features of the metrics that rank the top three in feature importance. The top three graphics describe the data features at the turning points where the situation turns from a disadvantage to an advantage, and the three graphics below describe the corresponding data features at the turning points where it changes from advantage to disadvantage.

Combining the previous analysis in this article and the definition of turning points, the turning points are a critical indicator for determining whether there is a fluctuation in momentum. Therefore, the differences in fluctuations are reflected in the differences in data features at turning points in different locations.

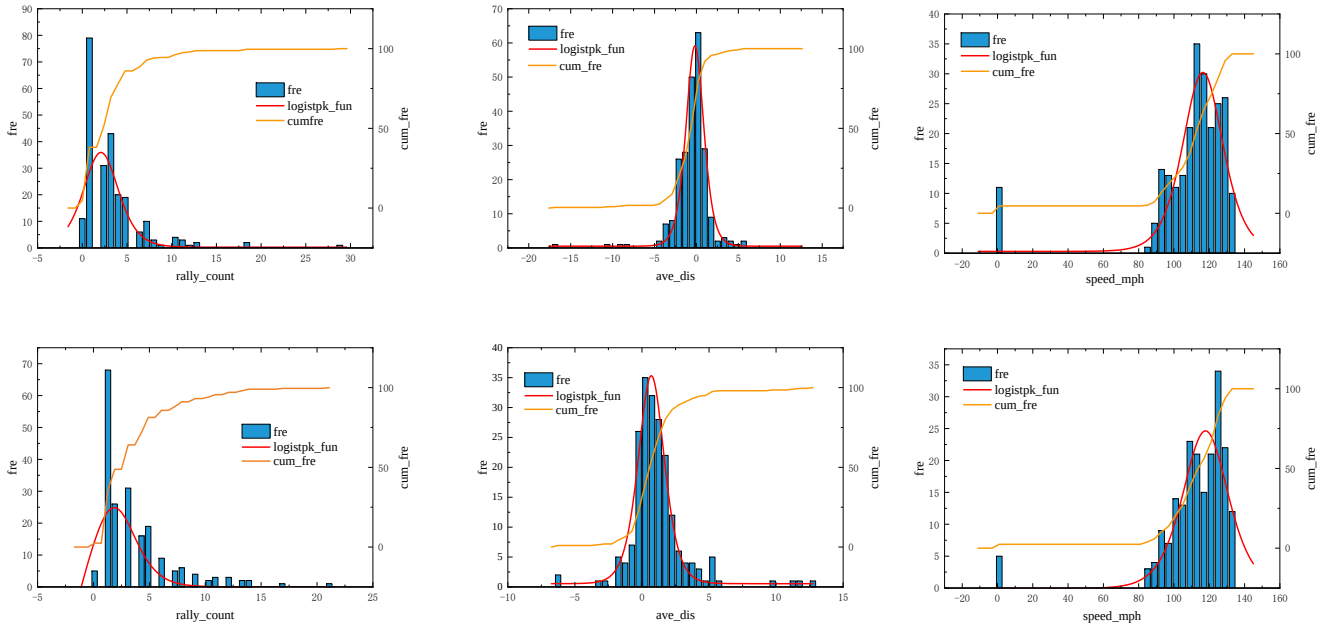

Figure 11: **Frequency Distribution of Data Characteristics for Key Indicators**

Through visual data analysis, we can find that for the number of strokes (rallycount), player1's turning points, whether in a favorable or unfavorable situation, are distributed around times when they serve or just after serving. However, by further digging into the data features at the serving points, we found that player1 gains advantage by scoring on serves when turning from a disadvantage to an advantage, while the opponent scores when player1's situation turns from being in an advantage to a disadvantage. Therefore, players should strive to score on their serves and prevent opponents from scoring on their own serves to seize the advantage.

For relative running distance per round, we find that situations where the disadvantaged party becomes the advantaged party often happens when their running distance each round is less than their opponent's. Conversely, instances where the advantaged party becomes the disadvantaged one often happen when the running distance per round is greater than the opponent's. Therefore, the disadvantaged party can regain the advantage by trying to keep their relative running distance per round below their opponent's. Similarly, to maintain an advantage, the advantaged party should try to minimize their own running distance as much as possible.

Similar to rally-count, for speed-mph, we find that whether player1 is the advantaged party or the disadvantaged party, the striking speed at their turning points is distributed around 120 mph. Therefore, when player1 is at a disadvantage, they should strive to turn the situation around by striking the ball at high speed. Conversely, when player1 has the advantage, they should be cautious to ensure they can effectively handle high-speed balls to prevent any errors that could turn the game around.

The distribution of feature importance percentages during different matches against different players can be used to formulate strategic guidelines for a player when facing different types of opponents. Here we take as an example the radar plot of feature importance when AC faces opponents 1701, 1401, and 1301. For opponent 1301 and 1701, they have a similar game, heavily influenced by the running distance. Therefore, AC can gain an advantage by trying to minimize their relative running distance. However, for the 1401 type opponent, they are more heavily influenced by the number of consecutive wins and the speed of the ball. Consequently, AC should prevent the opponent from getting consecutive wins and try to control the speed of the ball to defeat the opponent.

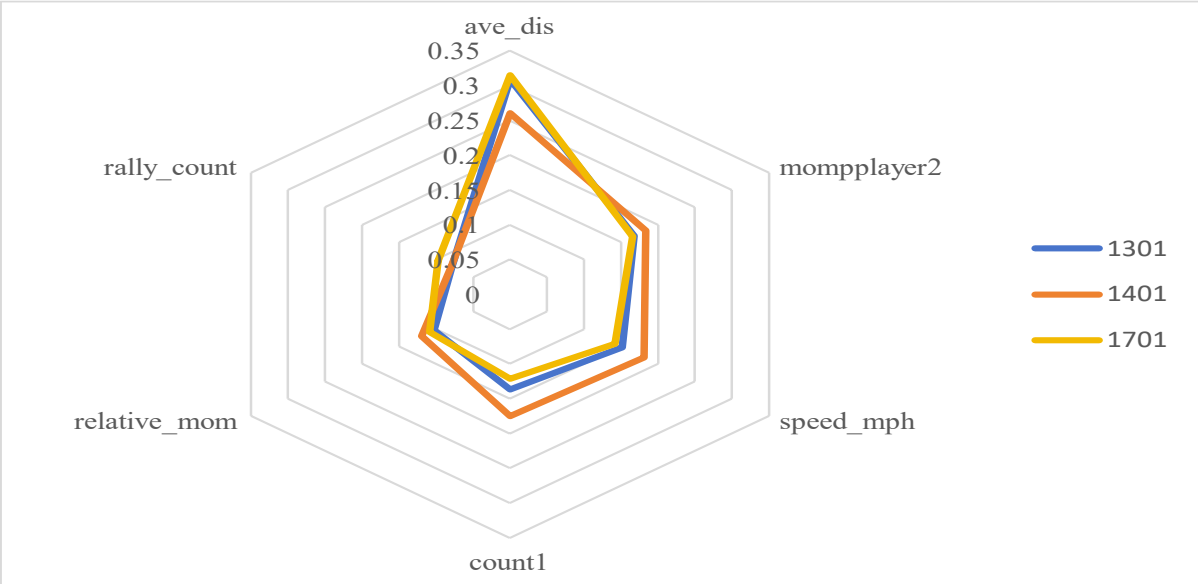

Figure 12: **Important indicators as a percentage of different competitions**

8 Testing model fit

8.1 Adaptability of the model to similar competitions

We use the model in the first question of the third question to simulate the prediction of turning points during the game for the four games 1301, 1401, 1501, and 1601 where paly1 are all AC, and test the data with the turning points in the real data to get the parameter values of the model's recall, precision, and F1, and plot the line graphs of the results of its four tests as follows.

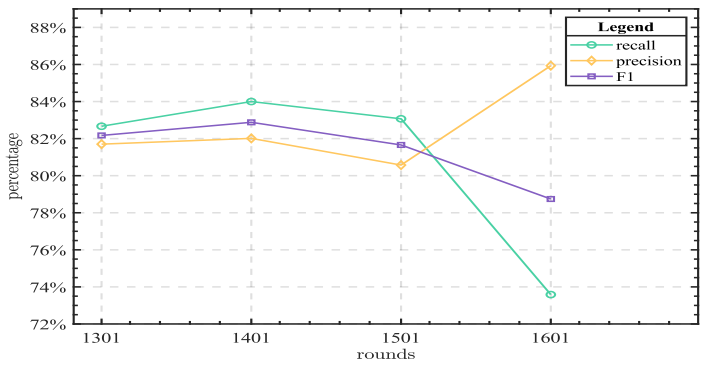

Figure 13: **Predicted results before improvements**

We found that except for the data of game 1601, our model predicts the turning points of the other three games better, which are all above 80%. For the 1601 game which is poorly predicted, according to our definition of turning point in Q3 i.e. turning point is one of the indicators that show the change of player's momentum, we consider that it may be because the model lacks the relevant data features that affect the momentum as analyzed in Q1, so we introduce it to optimize the model. That is, we define ace, winner-a, double, break-pt, break-pt-won, unf-err., double-fault, Server as the added data features, and re-train the model using 1701 as the training set of the model. And 1601 is used as the test set to test the effective degree of the optimized model to predict the turning point, and it is found that the parameter indicators such as the prediction accuracy of the optimized model have been significantly improved, and the following line graph of the detection effect after optimization is obtained.

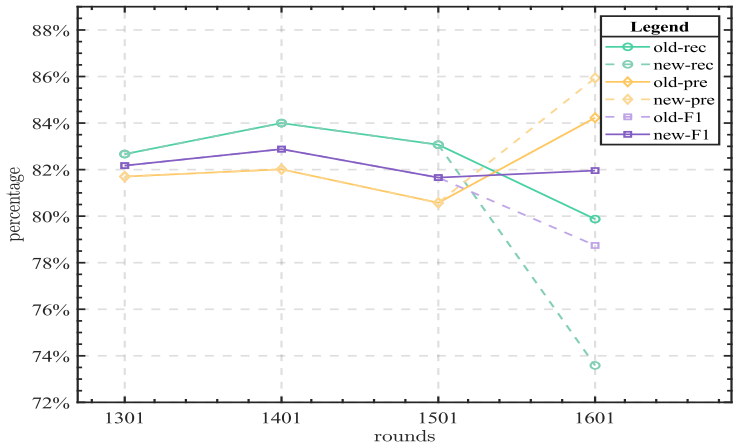

Figure 14: **Improved prediction results**

It can be seen that the model's prediction accuracy improves considerably with the addition of the new indicator factor。

8.2 Adaptability of the model to other types of competition

Introducing the 2020 Australian open women's tennis match data for testing and analyzing through the feature percentage bar chart, although there is a slight difference in the importance of the features, the indicators are still attributed to the category of tennis, and the model still shows a good accuracy in its prediction, which shows that the model has a good generalization to other tennis matches.

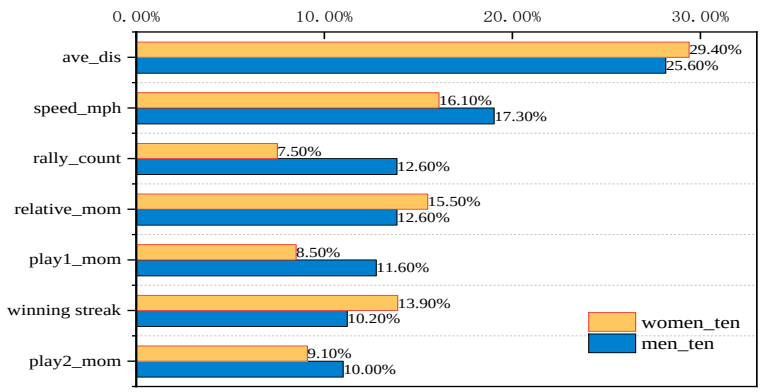

Figure 15: **omparison of Importance of Characterization Indicators**

Table 6: Women's Tennis Predictive Accuracy

	Accuracy	Recall	Precision	F1
Training Sets	1	1	1	1
Test set	0.817	0.817	0.895	0.854

In terms of other sports, first, there are too many tennis-specific features in our model, such as ACEs, break points, etc. Second, sports like ping pong differ significantly from tennis in certain feature metrics, such as running distance, number of serves, etc., on which the model strongly depends, thus the model has lower generality when dealing with other sports. If you want to improve its accuracy on them, you should consider reselecting and analyzing the feature indicators.

9 Sensitivity Analysis and Error Analysis

9.1 Error analysis for problem 2

In Problem 2, we attempted to show that the relative momentum of the whole game is correlated with the final outcome of the game, here we fitted the game wins and losses to a curve and tested the reliability of the claim that they are correlated by evaluating the distribution of the relative momentum around the curve.

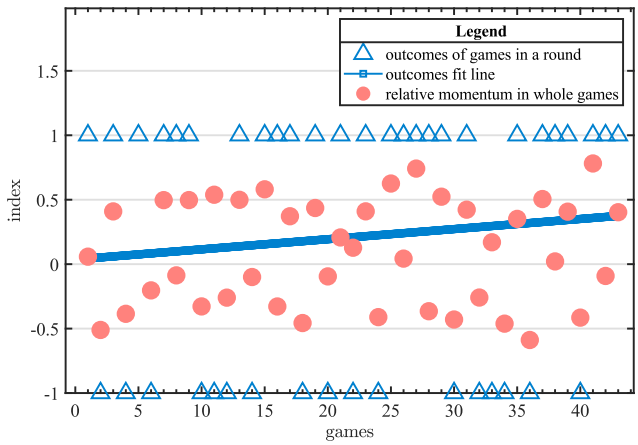

Figure 16: **Fitted Curve**

After fitting we get the curve

$$f(x) = p1 * x + p2 \tag{21}$$

where $p1 = 0.0079$, the lower and upper bounds on the confidence interval with 95% confidence level are -0.0169, 0.0326, respectively; $p2 = 0.0365$, the lower and upper bounds on the confidence interval with 95% confidence level are-0.5881,0.6612. The goodness-of-fit SSE is equal to40.7079. It is not difficult to imagine that these scatters are located within the region circled by the upper and lower bounds, proving that the correlation analysis error of Problem 2 is not large enough to be acceptable.

9.2 Sensitivity analysis for question 3

We add 1 Gaussian noise to only one set of feature metrics, speed-mph/ave-dis/rally-count, respectively, each time, while keeping the other feature metrics unchanged

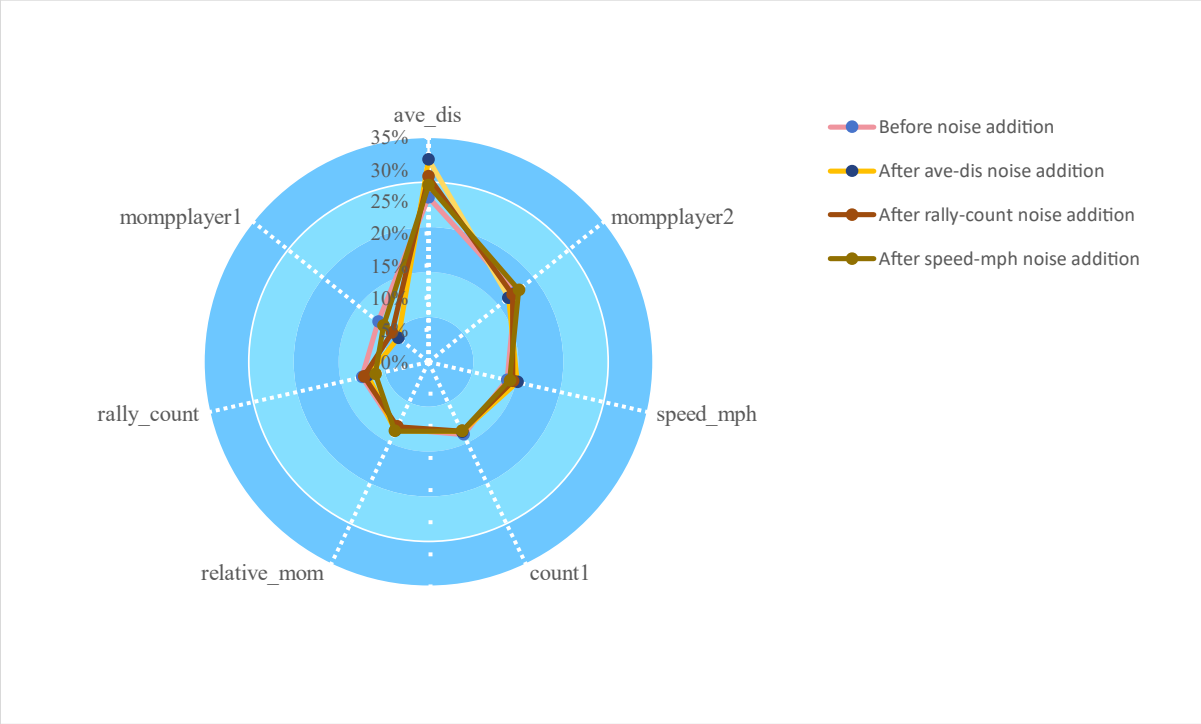

Figure 17: **Sensitivity Analysis Chart**

The intuitive results of the sensitivity analysis show that the GBDT classification model exhibits low sensitivity to changes in each of the important indicator features, indicating high stability.

10 Model Evaluation and Further Discussion

10.1 Strengths

1.The GDBT model iterates continuously during training, fitting in the negative gradient direction to ensure that each round of training can reduce the loss function as quickly as possible, accelerating convergence to a local or global optimum. This is beneficial for us to more accurately determine the turning points of the match, providing a solid basis for arranging tactics rationally for players.

2.Tennis matches are competitions between two individuals, where their difference determines the outcome. Innovatively, we consider the gap in certain data items between the two players as indicators to assess the situation of the match, reflecting the relative positions of both sides, rather than looking at each person in isolation.

3.The model reflects reality well. When training and testing with match data from Carlos Alcaraz vs. Novak Djokovic, as well as other matches, the accuracy generally exceeds 90%.

4.We considered not only direct factors but also indirect factors, enabling us to provide a richer set of winning strategies for players.

10.2 Weaknesses And Further Discussion

Weaknesses:

1.The absence of thorough feature engineering means the model's performance and generalization capability are relatively low.

2.Our suggestion to use inflection points to guide players focuses on preventing a shift from advantage to disadvantage but doesn't adequately address how players in a disadvantageous position can break through and gain the upper hand.

3.Despite our efforts to identify factors related to momentum and the outcomes of matches, it's possible that some influencing elements have been overlooked.

Further Discussion:

Consider merging GBDT with other models, such as Random Forest, Support Vector Machines, or Neural Networks, to achieve better performance and robustness. Additionally, employing regularization techniques (such as L1/L2 regularization) can help reduce model complexity and the risk of overfitting.

11 Conclusion

We assessed and predicted the performance of players using the momentum system, which showed strong consistency with the actual data. Having found that we can describe player performance through momentum, we assigned weights to multi-dimensional game data using entropy weight method and quantified momentum using the TOPSIS method. We innovatively proposed the concept of relative momentum and validated that it could better describe player performance as well as predict a player's scores and outcomes in the game.

We also put forward the concept of a "turning point", using the occurrence of turning points as an alternative to predicting turning points. We utilize this logic to predict the fluctuation of the game based on the state of the point before and that one. We've also determined the key factors influencing game fluctuations, providing a basis for players to formulate strategies.

Using the model we established, we made predictions on the provided game data, which showed high accuracy compared to the actual results. At the same time, we are also trying to introduce other minor variables to improve our model, with the aim to expand it for other types of tennis matches or even other ball sports.

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Letter

Dear Sir or Madam,

We are honored to introduce to you our analysis and modeling of the fluctuations in tennis matches and the role of "momentum". We believe our research results will help you better understand momentum, and assist in devising better match decisions and training plans for future match schedules, in order to improve players' performance.

Here are some of our analysis and forecast results.

Firstly, we determined the weight of factors affecting the match, where data from break points, ACEs, winning streaks, unforced errors, etc., have a larger weight. We quantified the so-called 'momentum' through these indices. We found an interesting data feature, although the number of hits and running distance, which were obtained in the entropy weight method, have a small weight in describing momentum (i.e., the performance of the athlete), they still impact the performance of the athletes and the trend of the match in subsequent tests. We will explain this data feature in the following analysis.

When we give a clear definition and calculation feature of momentum, through visual analysis, in the match flow chart, we can find the consistency of the winning streak and momentum trends, and the relative size of the momentum will affect the outcome of each game.

We validated our thoughts through spearman correlation analysis. The correlation coefficient of momentum and winning streak is 0.641, and the correlation coefficient of the cumulative sum of relative momentum per game with victory or defeat is 0.674. Overall, there is correlation.

Next, we define turning points, train the GBDT model, and analyze the fluctuations of the match and when the merits of the players switch. For match No. 1701, we can find interesting features on the turning points, and the features with smaller weights in the entropy weight method, such feature indices display a higher importance in the model we have established for turning points. We conjectured that although these indicators don't directly affect the performance of the players, they indirectly impact the match trend by changing the player's momentum. We notice that for players, whether in advantage or disadvantage, their turning points are located when serving or shortly after serving. But we further excavate serve point data features and find that the player switches from disadvantage to advantage by scoring serves, and switches from advantage to disadvantage when the opponent scores serves. For each round's relative running distance, we find that a disadvantaged side often becomes the advantage side when their running distance per round is less than that of their opponent, whereas the advantage side often becomes the disadvantaged side when their running distance per round is greater than their opponent's. For speed-mph, we find that no matter if the player is in advantage or disadvantage, the hitting speed at their turning point is distributed at 120 mph.

Lastly, we found that for the GBDT-based turning point prediction, we can better control the game by broadening the dimension of judging turning points. For similar or close sports, like women's tennis, the model can be applied directly. For sports with bigger differences, such as ping pong, the model's application capability is poor.

Coaches can assess the performance of the players by calculating their momentum and provide targeted suggestions based on the player's trend at different stages of the match. Regarding the number of shots, coaches should train players to strive for serving scores and prevent opponents from scoring serves to vie for advantage. Regarding the relative running distance of each round, coaches should train players to control their own running distance to be less than their opponents when they are at a disadvantage, and to reduce their running distance when they are at an advantage to maintain the stability of the match situation. For hitting speed, coaches should focus more on training players' ability to hit high-speed balls, to make winning shots that the opponent will not be able to react to in time.

The above are the suggestions and strategies our team provides for you as a coach. Thank you for your time in watching this, and we hope that our model and these conclusions can help you!

You can read the full text of our model if you would like to learn more detailed information.

Sincerely

Team 2424943

Report on Use of AI

1. OpenAI *ChatGPT* (Nov 5, 2023 version, ChatGPT-4)

Query1: <`matlab

```
function coeff = mySpearman(X, Y)
% This function is used to calculate the Spearman rank correlation coefficient.

% Input:
%     X: input numerical sequence
%     Y: input numerical sequence

% Output:
%     coeff: the correlation coefficient of the two input sequences X and Y

if length(X) ~= length(Y)
    error('The dimensions of the two numerical sequences are not equal.');
    return;
end

N = length(X); % Get the length of the sequence

Xrank = zeros(1, N); % Store the ranks of elements in X
Yrank = zeros(1, N); % Store the ranks of elements in Y

[~, Xorder] = sort(X); % Get the sorted indices of X
[~, Yorder] = sort(Y); % Get the sorted indices of Y

% Calculate the ranks in Xrank
for i = 1:N
    Xrank(Xorder(i)) = i;
end

% Calculate the ranks in Yrank
for i = 1:N
    Yrank(Yorder(i)) = i;
end

% Calculate the Spearman rank correlation coefficient using the ranks
fenzi = 6 * sum((Xrank - Yrank).^2);
fenmu = N * (N^2 - 1);
coeff = 1 - fenzi / fenmu;

end % The function mySpearman ends here
```

``

The code contains an error. The error lies in the incorrect calculation of ranks. To rectify the issue, the code needs to calculate the ranks based on the original data, rather than using the erroneous rank calculations.>

Output: < The corrected code is as follows:

```matlab
function coeff = mySpearman(X, Y)
% This function is used to calculate the Spearman rank correlation coefficient.

% Input:
%     X: input numerical sequence
%     Y: input numerical sequence

% Output:
%     coeff: the correlation coefficient of the two input sequences X and Y

if length(X) ~= length(Y)
    error('The dimensions of the two numerical sequences are not equal.');
    return;
end

N = length(X); % Get the length of the sequence

Xrank = zeros(1, N); % Store the ranks of elements in X
Yrank = zeros(1, N); % Store the ranks of elements in Y

[sortedX, Xorder] = sort(X); % Get the sorted X values and their indices
[sortedY, Yorder] = sort(Y); % Get the sorted Y values and their indices

% Calculate the ranks in Xrank
Xrank(Xorder) = 1:N;

% Calculate the ranks in Yrank
Yrank(Yorder) = 1:N;

% Calculate the Spearman rank correlation coefficient using the ranks
fenzi = 6 * sum((Xrank - Yrank).^2);
fenmu = N * (N^2 - 1);
coeff = 1 - fenzi / fenmu;

end % The function mySpearman ends here
```

```

```

The corrected code calculates the ranks correctly by first sorting the input sequences and then assigning ranks based on the sorted indices. This ensures that the ranks are based on the original data, resulting in an accurate calculation of the Spearman rank correlation coefficient.>